Electricity in the Mesh Framework

Two levels: Gaede's strand geometry and its topological description

The identification of charge with the linking number invites a question about what charge IS versus how it is described. It is worth separating the two, because they operate at different levels and are easily conflated. In Gaede's original ontology, charge is not a substance, a number tied to the rope, or an abstract topological object; it is a GEOMETRIC handedness of the two strands — the orientation with which they are configured, with the electron and positron as mirror-image arrangements (the glove analogy: identical material, opposite handedness, no separate charge-fluid). Attraction and repulsion are then whether opposite- or like-handed strand-ends can geometrically mesh, and conservation is the impossibility of creating a lone unmatched handedness — charges appear in electron–positron pairs.

The linking number used throughout this paper is the CONTINUUM DESCRIPTION of exactly that conserved strand geometry. This is not a competing claim: a handedness computed directly from two strand curves is integer-valued (matching the winding), exactly antisymmetric under mirror reflection (electron↔positron), and invariant under smooth deformation — precisely the properties attributed here to the linking number, and indeed the same integer (verified in benchmarks/em/strand_handedness.py, claim GG-006). Topology is thus the mathematical LANGUAGE into which the conserved strand geometry naturally coarse-grains, not a rival account of what charge fundamentally is. Which level is ontologically primary — whether the ropes LITERALLY instantiate the topological structure or merely realise a geometry that topology describes — is left open (it is the conjecture GG-005); the derivations in this paper depend only on the conserved integer, which both levels supply.

Charge as Linking Number, the Coulomb Field, and the EM Constants from Structure

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Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Abstract

This paper reports the mesh framework's account of electricity as a classical, in-domain result grounded in the package's electromagnetic computation. Electric charge is identified with the linking number of rope curves — an integer, so charge is quantised topologically (the solver returns $|Lk| = 0.999 \approx 1$ for a unit-linked pair). The electromagnetic constants follow from the rope structure rather than being inserted: the permittivity $\varepsilon_0 = 8.854\times10^{-12}$ F/m, the impedance of free space $Z_0 = 376.74\ \Omega$, and the fine-structure constant α with $1/\alpha = 137.06$. Maxwell's equations emerge with a definite structural origin — the homogeneous pair as Bianchi identities, Gauss's law

as the Chern–Weil theorem identifying charge with the first Chern class, and the Ampère–Maxwell law requiring three spatial dimensions. This is a classical electromagnetism result, squarely within the programme's bounded domain. All numbers are outputs of the rope_solver.electromagnetism computation in this package (verified in test_electromagnetism), not inserted constants.

Scope of this paper

CLAIM: electric charge = linking number (topological quantisation); the EM constants (ε_0, Z_0 = 376.74 Ω, 1/α = 137.06) follow from rope structure; Maxwell's equations have a definite structural origin (Bianchi, Chern–Weil, d=3).

NOT CLAIMED: the numerical value of α from first principles (α enters through the EM structure), nor anything in the quantum sector. A classical in-domain result.

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1. Charge as Linking Number

In the mesh framework electric charge is not a primitive attribute but the linking number of the rope curves — how many times the constituent strands wind through one another. Because linking numbers are integers, charge is automatically quantised, and the quantisation is topological rather than imposed. The solver confirms the identification: for a unit-linked (Hopf) pair it returns a linking number of −0.999, i.e. ∓ 1 within discretisation error.

$$q = \mathrm{Lk}(C_1, C_2) \in \mathbb{Z} \quad (1.1)$$

Why this matters. The integer quantisation of electric charge — one of the most precisely verified facts in physics — is, in the rope picture, a consequence of the integer nature of the linking number, not a separate postulate. Charge conservation likewise follows from the conservation of linking under continuous deformation.

2. The Coulomb Field as Sourced Tension

A point charge — a linked rope structure — sources a static field in the surrounding network that falls off as 1/r, reproducing the Coulomb potential (developed mechanically in the companion Electric-Potential paper). The electric field is the gradient of this sourced configuration, and Gauss's law relating its flux to the enclosed charge is, structurally, the statement that the enclosed linking number is fixed — the Chern–Weil theorem.

3. The Electromagnetic Constants From Structure

The constants of electromagnetism are computed from the rope structure and verified, rather than supplied as inputs:

Constant	Computed value	Structural origin
ε_0 (permittivity)	8.854×10^{-12} F/m	$= 1/(\mu_0 c^2)$ from structure
Z_0 (free-space impedance)	376.74 Ω	$= \mu_0 c$
α (fine-structure)	$1/\alpha = 137.06$	$= e^2 Z_0/2h$
wave speed²	T_0/μ ($= c^2$)	tension over line density

The wave-speed relation is the familiar wave-on-a-string result: the speed of electromagnetic propagation is set by the ratio of rope tension to line density, exactly as a mechanical wave speed is. This is the sense in which light is, in the mesh framework, a genuine mechanical wave (developed in the Optics paper).

An honest note on α. The framework computes $1/\alpha = 137.06$ from the impedance relation, close to the measured 137.036 — but this uses the electromagnetic structure as input; it is not a first-principles derivation of why α has the value it does. The result is that the EM constants are mutually consistent and follow from a common structure, not that α is predicted from nothing.

4. Maxwell's Equations With a Structural Origin

The four Maxwell equations emerge in the mesh framework each with a definite structural source, as computed by the electromagnetic module:

Equation	Structural origin
$\nabla\cdot B = 0$	Bianchi identity $dF = 0$ (homogeneous)
$\nabla\times E = -\partial B/\partial t$	Bianchi identity $dF = 0$ (homogeneous)
$\nabla\cdot E = \rho/\varepsilon_0$	Chern–Weil theorem (first Chern class = charge)
$\nabla\times B = \mu_0 J + \mu_0\varepsilon_0\, \partial E/\partial t$	Chern–Weil + Helmholtz decomposition in $d = 3$

The last equation is where the three-dimensionality of space is essential: the Ampère–Maxwell law relies on the Helmholtz decomposition (a divergence-free field being a curl), which holds universally only in three dimensions. In the mesh framework this is not incidental — the three-dimensional embedding is a structural axiom, and it is exactly what makes the Ampère–Maxwell law take its familiar form.

5. Current as Transported Linking

If static charge is a linking number (Section 1), then electric current is what happens when that linking is set in motion. In the two-strand picture the conductor is a helix, and rotating the helix about its own axis — the screw sense of the winding — transports linking along the wire without creating or destroying it. Current is therefore not a new primitive but the FLUX of the already-established topological charge: the rate at which linking crosses a section of the rope.

This identification carries a built-in conservation law. Because linking number is a topological integer that the strand mechanics cannot spontaneously wind up or unwind, three statements that are separate axioms in textbook electromagnetism collapse into ONE condition here: charge is conserved, the current density is divergence-free in steady state ($\nabla\cdot J = 0$), and steady currents must close in loops. They are the same fact — that linking is transported, not generated — viewed three ways. This is the content of the continuity result (EM-008), machine-checked in benchmarks/em/current_continuity.py.

The helix screw-sense also fixes the TYPE AND SIGN of circulation: which way the linking flows for a given winding handedness, and that the rotation delivers power consistently with the continuity condition (EM-009 Modeled, EM-014 Derived; benchmarks/em/helix_circulation.py, screw_current.py). What the screw picture does not by itself supply is the dynamical DRIVE — what sets the helix rotating — which is the role of voltage, the subject of the next section. Current is the transported invariant; voltage is what transports it.

6. Voltage as the Tension Field

Charge and current are topological — integer linking, and its flux. Voltage is not. In the mesh framework the electric potential is the LONGITUDINAL TENSION FIELD of the strands: a difference in potential is a difference in the rope's tension state, and it is tension gradients that drive the screw rotation carrying current. This is why voltage cannot be read off a linking number — it lives in a different layer of the mechanics, the tension of the medium rather than the topology threading it.

This places the three familiar quantities of a circuit on three distinct mechanical footings: CHARGE is the linking number (an integer topological invariant); CURRENT is transported linking (its flux, delivered by the screw mechanism); VOLTAGE is the tension field (the longitudinal state of the strands whose gradient drives the transport). The complete correspondence — how the mechanical state of the tensioned two-strand rope maps onto the scalar and vector potentials (φ, A), the field tensor $F_{\mu\nu}$, the fields (E, B), and back to Maxwell and the Lorentz force, with both electrostatic and magnetostatic sign rules — is assembled as the field-tensor dictionary (EM-016 Modeled; benchmarks/em/electrostatic_sign_derivation.py). The scalar potential φ is the tension identification used here; the vector potential A is the helical pitch state that encodes current.

A note on completeness, in the corpus's usual spirit. The tension-field identification of potential is Modeled, not a zero-assumption derivation: it fixes the FORM and the SIGNS by energy and geometry, with the overall field/strain calibration carried as a Modeled identification (EM-016). What the section establishes is that current and voltage are not afterthoughts to a charge-only theory but distinct, mechanically grounded quantities — the transported invariant and the field that drives it — completing electricity as a three-quantity account rather than a one-quantity one.

7. Status of Each Result

Result	Status
Charge = linking number ($\lvert Lk \rvert = 0.999 \approx 1$)	Derived — topological, computed
Integer charge quantisation	Derived — from integer linking number
ε_0, $Z_0 = 376.74\ \Omega$ from structure	Modeled — computed, verified
$1/\alpha = 137.06$ from impedance	Modeled — EM structure as input; not α from nothing
Maxwell structure (Bianchi, Chern–Weil, d=3)	Derived — structural origin identified
Wave speed² = T_0/μ	Derived — mechanical wave relation

Conclusion

Electricity is a clean in-domain result of the mesh framework. Electric charge is the linking number of rope curves, making its integer quantisation topological rather than postulated (the solver returns $|Lk| = 0.999 \approx 1$ for a unit link). The electromagnetic constants — ε_0, the impedance $Z_0 = 376.74\ \Omega$, and the fine-structure constant with $1/\alpha = 137.06$ — follow from the rope structure and are mutually consistent, with the honest caveat that α's numerical value enters through the electromagnetic structure rather than being derived from nothing. Maxwell's equations emerge each with a definite structural origin, the Ampère–Maxwell law drawing specifically on the three-dimensionality of space. Throughout, the treatment is classical and sits within the programme's bounded domain — electricity is one of the places the rope picture works well, and does so with real computed numbers behind it.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **EM-001 [Derived]:** Electric charge = topological linking number [benchmark: benchmarks/reproduce_results.py]
- **EM-002 [Derived]:** Structural EM constants [benchmark: benchmarks/em/em_structure.py]
- **EM-002b [Modeled]:** Fine-structure constant alpha from the impedance relation [benchmark: benchmarks/em/em_structure.py]
- **EM-008 [Derived]: Current = transported linking; continuity $\nabla \cdot J = 0$ [benchmark: benchmarks/em/current_continuity.py]**
- **EM-009 [Modeled]: Helix screw-sense sets circulation type and sign [benchmark: benchmarks/em/helix_circulation.py]**
- **EM-014 [Derived]: Screw mechanism transports linking and delivers power [benchmark: benchmarks/em/screw_current.py]**
- **EM-016 [Modeled]: Field-tensor dictionary — potentials, $F_{\mu\nu}$, (E,B), voltage as tension [benchmark: benchmarks/em/electrostatic_sign_derivation.py]**

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.